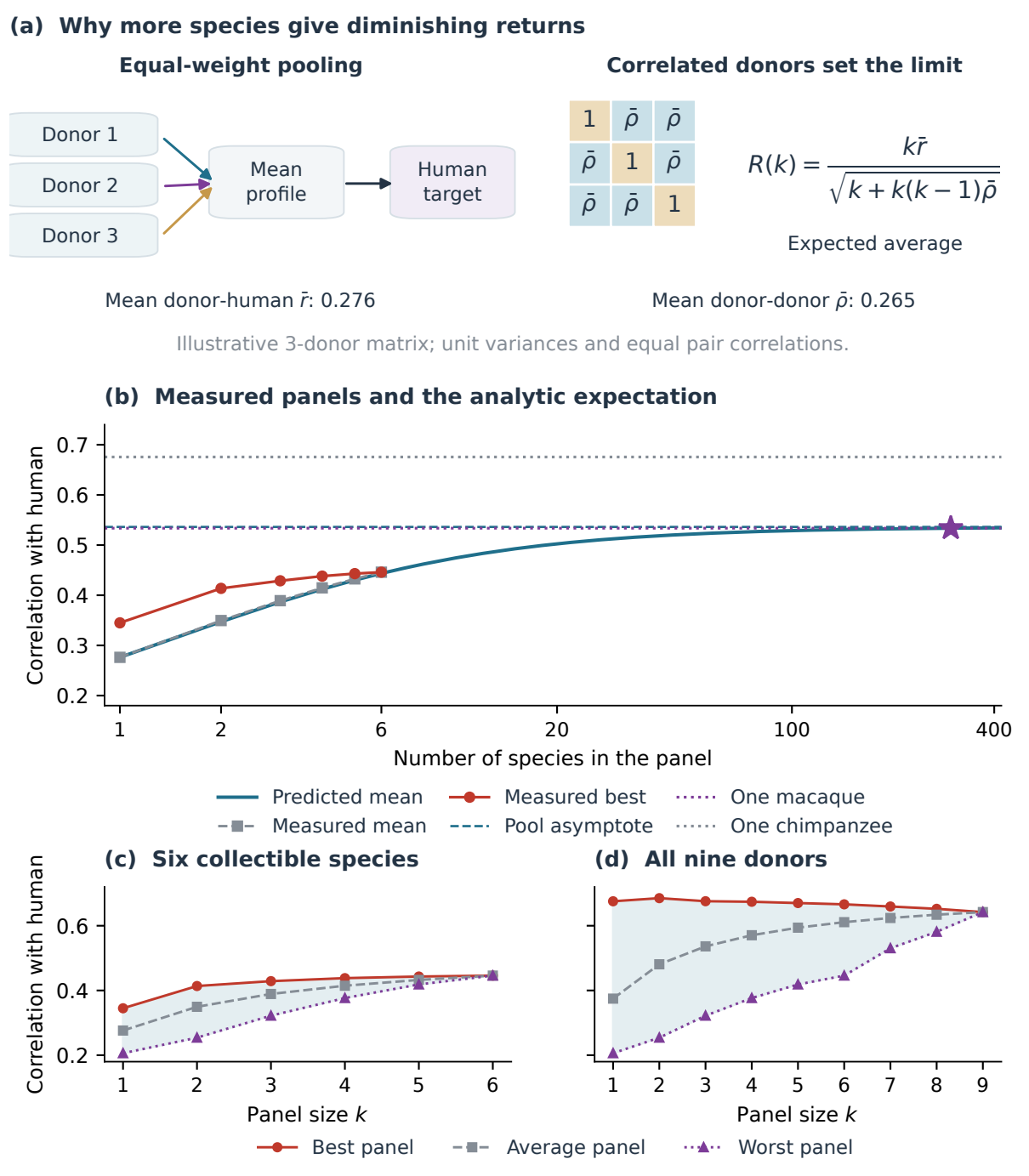
(a) Why more species give diminishing returns
Equal-weight pooling
Donor 1
Donor 2
Donor 3
Mean profile
Human target
Correlated donors set the limit
1 ρ̄ ρ̄
ρ̄ 1 ρ̄
ρ̄ ρ̄ 1
$R(k) = \frac{k\bar{r}}{\sqrt{k + k(k-1)\bar{\rho}}}$
Expected average
Mean donor-human r̄: 0.276
Mean donor-donor ρ̄: 0.265
Illustrative 3-donor matrix; unit variances and equal pair correlations.
(b) Measured panels and the analytic expectation
Correlation with human
0.7
0.6
0.5
0.4
0.3
0.2
1
2
6
20
100
400
Number of species in the panel
Predicted mean
Measured best
One macaque
Measured mean
Pool asymptote
One chimpanzee
(c) Six collectible species
Correlation with human
0.6
0.4
0.2
1
2
3
4
5
6
Panel size k
(d) All nine donors
1
2
3
4
5
6
7
8
9
Panel size k
Best panel
Average panel
Worst panel